Mp10 Modality Worklist

Installing, running and troubleshooting the DICOM worklist server

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1 About this guide

This guide is for whoever looks after the server: installing the Mp10 modality worklist, connecting imaging equipment to it, and working out what is wrong when a technologist says the worklist is empty.

A **modality worklist** is how a scanner finds out who its next patient is. Instead of a technologist typing the patient's name, date of birth and accession number at the console, the machine asks Mp10 and fills its own screen in. That is worth more than the keystrokes it saves: a study labelled by hand is a study that can be labelled wrongly, and a mistyped accession number is what puts an image in the wrong patient's file.

Two audiences, one document. The chapters on *what it is* and *day-to-day* are readable by anyone; the install and troubleshooting chapters assume you are administering the server.

2 How it fits together

A scanner does not speak HTTP. It speaks **DICOM**, over a TCP port, and asks a question called **C-FIND**. So the worklist is two pieces that meet in a directory on disk:

```
Mp10 orders (ADS)
     |
     |  AutoTasks service, every ~2 minutes
     v
the worklist directory        e.g. C:\Mp10Mwl\worklists
one small file per study      OR26-00041471.wl
     |
     |  DICOM worklist server (Orthanc), reading that directory
     v
C-FIND  <--------------------  the CT / ultrasound / X-ray room
```

Piece	What it is	Where it comes from
The generator	Part of the **AutoTasks** service. Decides which studies belong on the worklist and writes one file each	Installed with Mp10; switched on in the System Registry
The worklist directory	An ordinary folder. AutoTasks owns it completely	Created by the installer
The DICOM server	**Orthanc** with its worklist plugin, running as a Windows service	Installed once, from the Mp10 Modality Worklist bundle
The console	The Worklist page in Mp10 Web — read-only, for looking at what is scheduled	Part of Mp10 Web

The directory is the entire interface between the two halves. AutoTasks writes into it; Orthanc reads out of it; neither knows the other exists. That is what makes the failure described on page one of the troubleshooting chapter so common and so quiet — point them at different directories and each half works perfectly while the scanner shows nothing.

2.1 What is *not* here

This worklist replaces the worklist only. Images, the diagnostic viewer, dictation and report distribution are unchanged, and the HL7 orders Mp10 already sends to your imaging system keep flowing exactly as before.

The worklist server stores no images. It is configured to refuse them outright — it must never quietly become a second archive.

> **The worklist files themselves are patient data.** Each one carries a name, a date of birth and a medical record number. Keep the directory on local disk, off any share, and out of anything that copies folders to a cloud service. Orthanc's own web port is bound to the server's loopback address for the same reason: it is a health check, not a viewer.

3 Installing

3.1 Before you start

You need three answers. Get them from whoever configures the imaging equipment, because changing them afterwards means visiting every scanner:

	Bench default	What it is
AE title	`MP10MWL`	The name scanners call this server by. Up to 16 characters, no spaces
DICOM port	`4242`	The TCP port. Clinical sites usually want `104` or `11112`
Worklist directory	`C:\Mp10Mwl\worklists`	Where the files live. Local disk, not a network share

You also need, for each scanner: its **AE title**, its **IP address**, and — this one is easy to forget — **which fields it filters its query by**. Most filter by modality and date. Some also filter by *Scheduled Station AE Title* — "only my work" — which Mp10 supports and which is how you stop three CTs showing each other's patients, but which needs the mapping in *Sending each scanner its own work* filled in first. Find that out during the first scanner test, not during a clinic.

3.2 Install the DICOM server

On the server that will host the worklist, as **Administrator**:

```
powershell -NoProfile -ExecutionPolicy Bypass -File .\Install-Mwl.ps1
```

or, for a clinical install with the port and the scanners known:

```
.\Install-Mwl.ps1 -DicomPort 11112 `
                  -Modality "CT01,192.168.1.50" `
                  -Modality "MR01,192.168.1.51"
```

The installer needs no internet access: the Orthanc installer travels inside the bundle. It installs Orthanc, creates the worklist tree, installs **only** the worklist plugin, writes a locked-down configuration, opens the firewall port, restarts the service and then proves the server is answering before it says it is done.

With `-Modality` given, only those machines may query and their source address is checked. With none given, any host on the network may read the worklist — acceptable on a bench, worth tightening before go-live.

> **After any Orthanc upgrade, re-run `Install-Mwl.ps1`.** An upgrade restores the vendor's own configuration files, which turns worklist serving **off**. Nothing fails loudly when that happens: the service runs, the port answers, and every query returns nothing. `Test-Mwl.ps1` reports it in one line.

3.3 Switch on the Mp10 half

In **Admin10 → Global System Registry**, in this order:

1. Create the table.

Section	Entry	Set to
`GENERAL`	`MODALITY-WORKLIST`	`YES`

Then run a **file check** in Admin10. The table is created during that check and at no other time — setting this *after* a check has run does nothing until the next one.

Station routing needs **data dictionary 10.72 or later**. The same file check creates `mwlstations` and grants rights on it; until it has run, the generator logs `station routing is OFF` and publishes every study unrouted — safe, but Imaging Stations will not appear in Admin10.

2. Turn the generator on.

Section	Entry	Set to
`AUTOTASKS`	`EXPORT_MWL`	`YES`
`AUTOTASKS`	`MWL_SPOOL_PATH`	the worklist directory, e.g. `C:\Mp10Mwl\worklists`
`AUTOTASKS`	`MWL_WINDOW_DAYS`	`1` unless you have a reason

No restart is needed. **AutoTasks re-reads every setting at the start of each cycle**, so a change takes effect within a couple of minutes. (If you are watching for it, `AutoTasks.exe --mwl-run` applies it immediately.)

> These entries usually exist already, set to `NO` or blank. That is Mp10 writing down a question it asked, not somebody else's change. Your value always wins once you set it.

`MWL_SPOOL_PATH` must be **the same directory** the installer reported. Not a share pointing at it, not a different capitalisation of a different folder — the same directory.

3. Tell the watchdog where the server is.

Section	Entry	Set to
`AUTOTASKS`	`MWL_SCP_PORT`	the DICOM port the installer reported. `4242` unless you chose another
`AUTOTASKS`	`MWL_SCP_AET`	the server's AE title. `MP10MWL` unless you chose another

While the generator is on, AutoTasks asks the DICOM server every cycle whether it is answering. If it is not, twice in a row, AutoTasks restarts the Orthanc service once and sends a notice to the addresses in `ERRORREPORTS` / `NOTIFY_EMAIL` — see *The server is not answering*. Installed with the defaults, these two entries are already right and there is nothing to do. `AutoTasks.exe --mwl-probe` tells you in one line whether they are.

3.4 Fill in the modality mapping

A worklist entry has to say whether the study is a CT, an ultrasound or an X-ray. Mp10 takes that from the **revenue code** on the order: each revenue code carries the kind of imaging it represents.

Run the starter mapping once per practice:

```
Services\docs\seed-revcodes-modality.sql
```

It fills the standard revenue codes and deliberately leaves non-imaging ones blank. Then correct any local codes it did not cover.

Studies with no modality are not published at all. That is deliberate: DICOM has no way to say "kind of study unknown", and a blank one would match no scanner's query anyway. It is also the filter that keeps laboratory work and office visits off the imaging worklist. AutoTasks counts the unmapped ones every cycle, with examples — drive that number down to the point where what remains is genuinely non-imaging.

3.5 Prove it, in the right order

Test the DICOM half **before** any patient data is involved. Each step tells you something the next one cannot.

1. Is the server healthy?

```
powershell -File .\Test-Mwl.ps1 -ExpectedSpoolPath "C:\Mp10Mwl\worklists"
```

Everything green means the service is up, the configuration is Mp10's, the two halves agree on the directory, the port is listening and the firewall is open.

2. Can a machine see a study we publish?

```
AutoTasks.exe --mwl-selftest "C:\Mp10Mwl\worklists"
```

That writes one obviously-fake study — patient **MP10 SELFTEST / DO NOT SCAN**, a CT, scheduled today — using exactly the same machinery a real study goes through, with no database involved. Query the worklist from a scanner. If it appears, the whole DICOM path works: file format, server, port, firewall, scanner configuration.

The test file disappears on the next generator cycle, because AutoTasks owns that directory. Delete it by hand if you want it gone sooner.

3. Do real studies come through?

```
AutoTasks.exe --mwl-run
```

One worklist cycle and nothing else — no backups, no remittances, no e-mail. Safe to run in the middle of a clinic. It prints what it found and what it wrote. Then query from the scanner again.

If step 2 works and step 3 does not, the problem is in the **data**: nothing scheduled in the window, or revenue codes not mapped. If step 2 fails too, the problem is in the **DICOM half**, and step 1 says which part.

4 Sending each scanner its own work

With one CT, every CT study can go to every CT and nothing is wrong. With three, a technologist at one site should not be looking at another site's patients to find today's list.

DICOM's answer is **Scheduled Station AE Title**: the worklist entry says which station the study is booked on, and each scanner asks only for its own. Mp10 has no station field on an order — nobody chooses a scanner when the study is ordered — so the station is **derived**:

```
the encounter's location   (admit type: "Site A", "Site B" ...)
             +
the study's modality       (from the revenue code: CT, US, DX ...)
             |
             v
Tables -> Imaging Stations  ->  AE title the scanner calls itself
```

Nothing is typed at order entry, and adding a scanner is one row in the **Imaging Stations** screen, not a code change.

4.1 Where the screen is

Admin10 → the "Tables" panel on the left → Imaging Stations, between *Referring Phys* and *Fee Schedule*. It is a long list; the entry is about two-thirds of the way down.

> **If it is not there**, the feature is switched off in this dictionary. The link only appears where `GENERAL / MODALITY-WORKLIST` is `YES` — the same setting that decides whether the table exists at all, so a link to a missing table can never appear. Set it, run **Update Data Dictionary**, and restart Admin10: the panel is built once, when the program starts.

The screen also needs the dictionary at **version 10.72 or later**. Opening it against an older one says so rather than failing quietly.

4.2 The rows

Field	What it is
Location	Must match the encounter type used for that site. Spelling matters, capitalisation does not
Modality	`CT`, `MR`, `DX`, `US`, `MG` — the same value the revenue code maps to
AE Title	What the scanner calls itself when it asks for its worklist. This is the field that does the routing. **Capitalisation matters here**: `MxView` and `MXVIEW` are two different stations
Station Name	Free label, shown on some consoles
Notes	For people, not for DICOM — room, vendor, whoever to call
Inactive	Tick to stop routing to it without deleting the row

One row per location and modality. **Two X-ray rooms in the same building share a worklist** — that is Phase 1 as designed, not a fault. Splitting them needs a finer key than location and modality (body part is the intended next step).

4.2.1 Where each value comes from

Location	The encounter types you already use. If encounters are opened as `Site A`, `Site B` and `Site C`, those are the three values — typed exactly as they appear on the encounter, not abbreviated
Modality	Whatever the revenue codes map to. If no revenue code maps to `MR`, a station for `MR` will never be used, and the screen says so when you save
AE Title	**From the scanner, not from you.** It is the name the device calls *itself* in its worklist query — read it off the machine's DICOM configuration, or ask whoever installed it. Up to 16 characters, no spaces, and **typed exactly as the scanner has it, capitals and lower case included** — the DICOM server compares the two letter for letter. Do not rename a scanner to suit this table: its AE title is usually the name the image archive already knows it by. Getting this wrong is invisible: the row saves, looks right, and routes nothing

4.2.2 A worked example

Three buildings, one CT and one X-ray room in each:

Location	Modality	AE Title	Station Name	Notes
Site A	CT	`SITEA_CT`	Site A CT	Siemens, room 2
Site A	DX	`SITEA_DX`	Site A X-ray	
Site B	CT	`SITEB_CT`	Site B CT	
Site B	DX	`SITEB_DX`	Site B X-ray	
Site C	CT	`SITEC_CT`	Site C CT	
Site C	DX	`SITEC_DX`	Site C X-ray	

Six rows, and a CT study opened at Site B now goes to `SITEB_CT` and to no other machine. Substitute your own encounter types and the AE titles the devices actually use.

4.2.3 What the screen checks

It **refuses** a row with no Location, no Modality or no AE Title (a row with no AE title routes nothing — it exists only to be misread later), an AE title containing a space, and a second row for a location and modality that already has one.

It **asks** — rather than refuses — when the Location matches no encounter type or the Modality matches no revenue code, because a site may map a scanner before its first study goes through it. A typo produces exactly that question, so read it rather than clicking past it.

Location and Modality **cannot be edited afterwards**: changing either is indistinguishable from adding the station somewhere else, and would leave the old pairing routed to nothing. Delete the row and add it again.

4.3 Fill it in before you switch a scanner to station filtering

The order matters, for the reason in the troubleshooting chapter: a scanner that filters by station sees nothing for any location and modality you have not mapped.

1. Add a row for every (location, modality) that scanner will ask for.
2. Watch `AutoTasks.out` — `published with NO station` should not name that scanner's pairs any more. `AutoTasks.exe --mwl-run` gives the answer in seconds rather than at the next cycle.
3. Prove it before real work depends on it:

   ```
   AutoTasks.exe --mwl-selftest c:\Mp10Mwl\worklists CT SITEB_CT
   ```

   That publishes one obviously-fake study booked on `SITEB_CT`. The scanner configured as `SITEB_CT` should see it; the others should not.
4. Then switch the scanner to filter by station AE title.

Reverse the order and the symptom is an empty worklist on a machine that was working ten minutes ago.

4.4 What it does not change

A scanner that filters by **modality and date** is unaffected by any of this. It sees every study of its modality, mapped or not, exactly as before. You can run a mixed estate — some devices station-filtered, some not — and the only rule is the one above: any device that filters by station needs its pairs mapped.

4.5 When the scanner will not filter: routing by caller

Some consoles cannot be told to ask for their own station. A Philips Brilliance CT, for instance, asks for "all CT, today and tomorrow" and nothing else, so with three CTs in three buildings each of them lists all three buildings' work, however carefully the Imaging Stations table is filled in.

The server can do the filtering for them. Installed with `-FilterByCallingAet`, a query that names no station is answered with the entries booked on the station **named like the scanner itself** — the AE title it called us from — and nothing else. A query that does name a station is answered as asked.

```
.\Install-Mwl.ps1 -FilterByCallingAet
```

This changes the rule for **every** scanner using the server, not just the one you had in mind:

- A scanner with **no Imaging Stations row** gets an empty worklist, because an entry with no station no longer answers an unfiltered query.
- A row whose AE title is **not exactly** what the scanner sends — `MxView` is not `MXVIEW` — is the same as no row. The capture script in the troubleshooting chapter shows the exact name.
- A scanner that used to see several locations' work now sees only its own.

So before switching it on, make sure every machine that queries this server has its row, and type each AE title the way the machine sends it. `Test-Mwl.ps1` says whether routing by caller is on.

5 Day to day

5.1 Where things are

Imaging stations	Admin10 -> Tables -> **Imaging Stations**
Worklist files	`C:\Mp10Mwl\worklists\<order number>.wl`
AutoTasks log	`AutoTasks.out`, beside `AutoTasks.exe`
Orthanc logs	`C:\Program Files\Orthanc Server\Logs`
Orthanc configuration	`C:\Program Files\Orthanc Server\Configuration\orthanc.json` and `worklists.json`
Health check	`http://127.0.0.1:8042/system` on the server itself
What is published right now	`http://127.0.0.1:8042/worklists` — see *Seeing exactly what a scanner would get*

5.2 Which studies appear

An order is on the worklist when **all** of these are true:

- its encounter is **not** inactive,
- its encounter is **not** closed,
- its start date is within `MWL_WINDOW_DAYS` days either side of today (the encounter's admission date is used when the order has no start date of its own), and
- its revenue code maps to a modality.

Nothing is ticked per order. An order that meets those tests is on the worklist; one that stops meeting them comes off it, and its file is deleted within a couple of minutes.

Whether a given scanner then *sees* it is a second question, answered by *Sending each scanner its own work*.

5.3 A finished study stays on the list

Nothing tells Mp10 that a scan has been performed — the equipment reports that to the PACS, not to the worklist — so a study that has already been done remains on the worklist until its date falls outside the window, usually the next day. That is normal, and every modality worklist behaves this way; technologists work from the list and ignore what they have already done.

If it is a nuisance, the answer is a narrower `MWL_WINDOW_DAYS` (`0` = today only), not a change of procedure.

5.4 Cancelling is not deleting

When a study falls off the worklist its file is removed but its **entry is kept and marked cancelled**. Reopen the encounter, or move the date back into the window, and the *same* entry returns with the *same* study identifier — so a study is never split into two half-studies in your image archive.

Deleting the order itself is the one thing that removes the entry outright, which is as it should be.

5.5 What a healthy log looks like

In `AutoTasks.out`, every couple of minutes:

```
WriteWorklistFiles: active set size      488
WriteWorklistFiles: inserted 0  re-activated 0  unchanged 488
WriteWorklistFiles: rows with a rev_code but no revcodes.Modality mapped 56  example rev_codes 0320,035
WriteWorklistFiles: cancelled 0
```

```
EmitWorklistFiles: written 0  unchanged 432  purged 0  no modality (not emitted) 56  write failures 0
```

Read it as: 488 orders qualify, nothing changed this cycle, 432 studies are published to the machines, and 56 are held back for want of a modality — that last number is the one with work behind it.

`written` is high on the first cycle after a change and `0` the rest of the time. `write failures` should always be `0`.

Two more lines appear once station routing is in play:

```
EmitWorklistFiles: published with NO station -- invisible to any scanner that
filters by station AE title   135   example location/modality   SiteC/US SiteA/DX
EmitWorklistFiles: station routing is OFF -- no mwlstations, or it cannot be read
```

The first counts studies that no Imaging Stations row matched, and names examples — those are the rows to add. The second means the table is absent or unreadable: expected before the 10.72 file check has run, worth investigating after it.

5.6 Turning it off

Setting `AUTOTASKS / EXPORT_MWL` to `NO` stops the generator — and **leaves the files it has already written**. Orthanc goes on serving them, so the scanners keep seeing a worklist that has quietly stopped being true, which is worse than no worklist at all.

To stop publishing properly:

1. set `EXPORT_MWL` to `NO`,
2. wait one cycle, then **delete the `.wl` files** in the worklist directory.

To stop one *site or modality* rather than everything, tick **Inactive** on its Imaging Stations row instead; the generator keeps publishing, and those studies simply stop carrying a station.

5.7 Seeing exactly what a scanner would get

The log says how many studies were published. This says **what is in them**, decoded, without a scanner or any DICOM tooling. On the server itself:

`http://127.0.0.1:8042/worklists`	every published entry
`http://127.0.0.1:8042/worklists/<order number>`	just that one, e.g. `.../worklists/OR26-00046110`

The entry id **is** the Mp10 order number, so a question that starts "is order OR26-00046110 on the worklist, and does it say the right thing?" is one URL.


```
{
   "ID" : "OR26-00046110",
   "Tags" : {
      "AccessionNumber" : "EN26-00281843",
      "PatientID" : "00100947",
      "PatientName" : "PEREZ ALVIRA^INGRID",
      "ReferringPhysicianName" : "SALGUEIRO^JESUS M",
      "ScheduledProcedureStepSequence" : [ {
         "Modality" : "CT",
         "ScheduledProcedureStepStartDate" : "20260826",
         "ScheduledStationAETitle" : "SITEB_CT",
         "ScheduledStationName" : "Site B CT"
      } ],
      "StudyInstanceUID" : "2.25.155399574729555749339428222821724285 5"
```

```
  }
}
```

What to look at, in the order it usually matters:

- `ScheduledStationAETitle` — the routing answer. **Absent** means no Imaging Stations row matched, so a station-filtering scanner will not see this study.
- `Modality` — must be exactly what the scanner asks for. A study with no modality is never published at all, so if it is here, it has one.
- `ScheduledProcedureStepStartDate` — the day the scanner has to ask for.
- `PatientName` and `AccessionNumber` — what the technologist will see, and what ties the images back to the encounter.

Three things to know about this page:

> It is bound to the server's **loopback address**, so it only opens on the server itself — not from a workstation, by design. It shows **patient data**, so treat the browser window as you would any clinical screen. And it is **not** the worklist users are meant to read: that is the Worklist page in Mp10 Web.

If it answers **HTTP 400** instead of a list, something in the worklist directory is not readable as DICOM — see the troubleshooting chapter. Scanners are unaffected in that state; only this listing breaks.

5.8 Older equipment that "needs its own worklist server"

Some legacy devices — older ultrasound systems in particular — are said to need a worklist server of their own. Usually they do not. What they have is a fixed setting you cannot change on the device, and the mismatch shows up at the console as *cannot connect*, which reads like an incompatibility rather than a configuration difference.

Three fixed settings account for most of it:

The device insists on	What to do
A particular Called AE title it will not let you edit	Install with `-AcceptAnyCalledAet`. The server then answers whatever a device calls it. Verified against devices calling `PHILIPS_MWL`, `WORKLIST` and `HDI5000`
Port 104, the standard DICOM port	`-DicomPort 104`. Nothing else changes
Querying with almost no filters — no date, sometimes not even a modality	Already handled. This server answers a query with an empty Scheduled Procedure Step, or none at all, by returning everything

```
.\Install-Mwl.ps1 -DicomPort 104 -AcceptAnyCalledAet
```

> `-AcceptAnyCalledAet` is a real loosening: any host that can reach the port is answered, whatever it calls itself. Pair it with `-Modality`, which restricts by source address, and prefer changing the device where the device allows it.

Before assuming a device cannot do this at all, two checks settle it in minutes:

1. **Point it at this server and watch what it sends.** On the server run `Capture-MwlQuery.ps1` from the bundle: it turns Orthanc's log to verbose, prints every connection as it arrives — what the device called itself and called us, the query it sent, how many entries matched — and puts the log level back when you press Ctrl+C. A device that never appears is pointed at the wrong address or port; one that is *rejected* is calling us by the wrong name; one that is answered with n entries and still shows nothing is discarding them on its own side.

2. **Read the vendor's DICOM conformance statement** for that model and software version. It states plainly whether *Modality Worklist Information Model — FIND* is supported as an SCU, and which keys the device sends. If worklist is absent from that document, the device genuinely cannot do it and no server-side change will help.

If it turns out the device does not speak worklist at all, the only routes are a software option or upgrade from the vendor, or the interface box the site is presumably already running — and that is worth knowing definitely rather than by inference.

5.9 Adding a scanner

1. On the scanner, configure the worklist source: **called AE title**, **host** and **port** exactly as `Test-Mwl.ps1` reports them. Its own AE title can be anything, but write it down **exactly, capitals included** — it goes into Imaging Stations letter for letter. If in doubt, `Capture-MwlQuery.ps1` on the server prints the name the scanner really sends.
2. If the server was installed with `-Modality`, re-run `Install-Mwl.ps1` with the new scanner added to the list — otherwise it will be refused at association, which the scanner usually reports as "cannot connect".
3. Use `--mwl-selftest` (above) for the first query rather than a real patient.

5.10 Philips iU22 and iE33, software 6.x

The commonest "legacy ultrasound" at a site, and the machine most often given as the reason for keeping a second worklist server. It does not need one. Its conformance statement (iU22/iE33 R6.0.X.X, document 000168000000037) describes an ordinary Modality Worklist SCU: Implicit and Explicit VR Little Endian, ISO_IR 100 for characters — which is exactly what Mp10 publishes — and an automatic query of **modality plus today's date**, which every published entry answers.

Three of its habits are worth knowing before configuring it:

- It **abandons the whole update** on any response status but success or pending. There is no partial worklist and no retry queue, so if the server is down the console says *query failed* and keeps the list it already had. That is the machine working correctly, not a fault to chase.
- It queries **the one worklist device selected** on the Device Selection page. Selecting ours is what stops it reading the old server.
- Its query has three **optional** extra filters — AE Title, Station Name, System Location. Tick only AE Title, for the reason in the warning below.

5.10.1 On the scanner

1. Add the device. `Setups > Print/Network > Global Config > Devices > New Device`

Field	Value
Device Type	**DICOM Worklist Server**
Device Name	anything — an alias for the console's own lists
AE Title	the server's called AE title, `MP10MWL` unless it was installed with another. Exact, or the association is refused
IP Address	the worklist server
Port	`104`, or whatever `-DicomPort` the server was installed with

Press **Verify** before going any further. That is a C-ECHO, and it settles network, firewall, port and AE title in one click.

2. Point it at us. `Setups > Print/Network > Device Selection > Worklist`

Select the new device. **This is the cutover.** Leave the previous worklist device defined but unselected, so going back is one click rather than a re-typed configuration.

In the **Define Query** section, tick **AE Title** only; modality stays `US`. Set the automatic update interval (15–120 minutes) and/or on End Exam to suit the department.

3. Change nothing on `Global Config > System`. The AE Title there — the name the scanner calls *itself* — is the value the Imaging Stations row routes on. Changing it empties the worklist, and the row still looks right.

> **Do not tick Station Name or System Location.** They buy nothing that AE Title does not already do, and AE Title is the field you control from Imaging Stations. System Location is worse than redundant: the server matches it **case-sensitively** (measured), and the value in the worklist entry is the encounter's location exactly as it is spelled in the **Admission Types** table — so it would have to be typed on every scanner character for character, and re-typed on all of them if that table is ever re-worded. Ticking either one also hides the `--mwl-selftest` entry, which carries labels of its own.

5.10.2 Proving it

Prove it, in the right order applies unchanged. On this machine the steps land as:

Do this	It tells you
`Test-Mwl.ps1 -ExpectedSpoolPath <dir>`	the server is healthy and both halves agree on the directory
Verify on the scanner's new device entry	network, port, called AE title
`AutoTasks.exe --mwl-selftest <dir> US <the scanner's own AE title>`, then **Update Worklist**	expect `MP10 SELFTEST / DO NOT SCAN`. Proves the file format and station routing with no patient data involved
`AutoTasks.exe --mwl-run`, then open `http://127.0.0.1:8042/worklists/<a US order number>` on the server	confirms `ScheduledStationAETitle` is present. If it is absent, no Imaging Stations row matched, and nothing will appear on the scanner
Update Worklist again	real patients, right names, accession = encounter number
Scan one patient from the list and find the study in the PACS	the acceptance test. Mp10 issues the Study Instance UID now, so this is the step that proves the loop closes

If this scanner's own AE title or port could not be changed — on this model both can — the fallbacks under *Older equipment that "needs its own worklist server"* apply unchanged.

5.11 Philips Brilliance CT

Set up at a site in one afternoon; these are the things that cost time.

- **Its own AE title is mixed case by default: `MxView`.** The Imaging Stations row must say `MxView`, not `MXVIEW` — the server matches it letter for letter. Do not rename the scanner to suit the table; the image archive already knows it by that name.
- **It does not filter by its own station.** Its worklist query is "all studies of my modality, today and tomorrow", and nothing else. With several CTs in several buildings, each lists all of them — however correct the Imaging Stations table is. Install the server with `-FilterByCallingAet` (see *When the scanner will not filter: routing by caller*) and it gets its own site's work.

- **Where the settings live.** Network → a *Node* with our server's IP → under it a *Device* of type HisRis with the called AE title, *Worklist* ticked, and the port under *Expert Mode*. The port is not on the first screen; left at the default it is 104, and nothing answers there.
- **Press Ok on every open dialog, then restart the DICOM services.** The console keeps running the last committed configuration. At one site the screen showed the right server name while the scanner was still sending a mistyped one, and the server rejected every query. The screen looked correct; the capture script showed the truth.
- A query that reaches the server and is answered shows on the console as the *Scheduled HIS/RIS* list. "0 entries" with no error usually means the query never reached us: wrong port, uncommitted dialogs, or the old server still selected.

6 Troubleshooting

Almost every worklist problem is one of two questions, and telling them apart is most of the work:

Is nothing being published, or is nothing getting through?

`Test-Mwl.ps1` answers the second. `AutoTasks.exe --mwl-run` answers the first. Run both before changing anything.

6.1 The scanner's worklist is empty

Work down this list; each step rules something out.

#	Check	How	If it fails
1	Are files being written?	Look in the worklist directory	Go to *Nothing is being published*
2	Do the two halves agree on the directory?	`Test-Mwl.ps1 -ExpectedSpoolPath <the Mp10 setting>`	Fix `MWL_SPOOL_PATH`, or re-run `Install-Mwl.ps1` with `-SpoolPath`
3	Is Orthanc actually serving?	`Test-Mwl.ps1` — "Worklist plugin is serving *n* entries", or open `http://127.0.0.1:8042/worklists` and read one	Go to *The server is not answering*
4	Does the scanner reach it?	`--mwl-selftest`, then query from the scanner	Go to *The scanner cannot connect*
5	Does the scanner filter by something we do not publish?	Ask it to query with no filters, or by date only	See *The scanner filters by station*

6.2 Nothing is being published

Symptom in `AutoTasks.out`	Cause
No `WriteWorklistFiles` lines at all	`AUTOTASKS / EXPORT_MWL` is not `YES`, or the service was not restarted after it was set
`no .wl written -- AUTOTASKS/MWL_SPOOL_PATH is empty`	The path setting is blank
`ABORTED -- AUTOTASKS/MWL_SPOOL_PATH does not exist`	The path is wrong, or the directory was deleted. AutoTasks will not create it: a mistyped path that created itself would look like it was working
`active set size 0` with orders on today's schedule	The encounters are closed or inactive, or the dates fall outside `MWL_WINDOW_DAYS`
`no modality (not emitted)` accounts for everything	The revenue-code mapping is not filled in. Nothing at all can be published until at least one code maps
`write failures` is not `0`	Permissions on the worklist directory, or the disk is full

If the table itself is missing, `GENERAL / MODALITY-WORKLIST` was set to `YES` after the last file check rather than before. Run the check again.

A file check does not complain when a table cannot be created. If `modalitywl` is still missing after a check that appeared to finish normally, report it rather than re-running it repeatedly.

6.3 The server is not answering

AutoTasks watches for this itself. Every cycle, while the generator is on, it opens and closes a DICOM connection to the server on `MWL_SCP_PORT`. Nothing is logged while the server answers. When it does not:

What AutoTasks finds	What it does
No answer, once	Logs it and waits for the next cycle — the server may simply be starting
No answer, twice in a row	Restarts the Orthanc service (ending its processes if it will not stop), waits for the port, and mails **"worklist server stopped answering and was restarted"** — or **"worklist server is DOWN"** if that did not bring it back
Still no answer on later cycles	Logs it. **It does not restart again during the same outage.** One reminder a day is mailed
It answers again	Mails **"worklist server is answering again"**
A DICOM server answers but refuses the connection	`MWL_SCP_PORT` or `MWL_SCP_AET` does not match the server. Nothing is restarted; one notice a day says so. Correct the entry — `Test-Mwl.ps1` prints the right values

Notices go to `ERRORREPORTS` / `NOTIFY_EMAIL`. With that entry blank, or no outgoing mail server configured, the restart still happens and the AutoTasks log is the only record. `MWL_SCP_RESTART = NO` keeps the notices and leaves the restarting to you; `MWL_SCP_PORT = 0` switches the whole check off.

`AutoTasks.exe --mwl-watchdog` runs the same check on demand, **including the restart and the notice**. `AutoTasks.exe --mwl-probe` only asks.

A restart puts the server back; it does not say why it stopped. That is in the server's log from before the restart, and the table below is how to read it.

Symptom	Cause
`Test-Mwl.ps1` says orthanc.json is **not** the Mp10 configuration	Orthanc was upgraded. Re-run `Install-Mwl.ps1`
The service is "Running" but the HTTP port does not answer	Orthanc itself died while its service wrapper stayed up — a healthy server shows two processes, `OrthancService` and `Orthanc`, and this one shows only the first. The reason is in the last lines of the newest file in `C:\Program Files\Orthanc Server\Logs`. Restart the service; if stopping hangs, end `OrthancService.exe` and start it
That log ends with *"The specified path does not point to a regular file: …<order number>.wl"*	The worklist plugin's housekeeper met a file the generator had just removed, and Orthanc stopped. Installations made before this was known have the housekeeper on. `Test-Mwl.ps1` says so; re-running `Install-Mwl.ps1` turns it off (`DeleteWorklistsOnStableStudy: false` in `worklists.json`)

Symptom	Cause
The log says a setting is *"defined in 2 different configuration files"*	An extra `.json` was added to Orthanc's configuration directory. Orthanc refuses to start rather than guess. Remove it and re-run `Install-Mwl.ps1`
`Worklist server is disabled by the configuration file`	`worklists.json` has `Enable: false` — again, usually an upgrade
Service start type is not Automatic	It will not come back after a reboot. `Install-Mwl.ps1` sets this
`Test-Mwl.ps1` warns **Orthanc cannot list the directory (HTTP 400)**	Something in the worklist directory is not readable as DICOM — usually a zero-length file, or something the generator did not write. **Scanners are unaffected**: a C-FIND skips the bad file and answers normally. Find it and delete it
`Test-Mwl.ps1` says *n* files are **not being served**	Same cause, seen from the other side: Orthanc parsed fewer files than are on disk. The warning prints the two commands that turn on verbose logging, which names the file

6.4 The scanner cannot connect

Symptom at the scanner	Cause
Called AE Title Not Recognized	The scanner is calling the wrong AE title. It must match exactly, including case
Association rejected, or refused	The server was installed with `-Modality` and this scanner is not on the list, or it is querying from a different IP than the one recorded
Times out with no response	Firewall, or the wrong port. `Test-Mwl.ps1` confirms the port is listening and the rule is enabled — from there it is the network between the two
Connects, returns nothing, no error	It is reaching the server. This is now a data or matching problem, not a connection one
Returns nothing, and the device's clock is wrong	The scanner asks for *its* today. If its date differs from the server's, the dates do not match and nothing comes back. Check the clock on the device before looking anywhere else

6.5 A station-filtering scanner sees nothing

Some equipment queries by **Scheduled Station AE Title** — "only show me work booked for *this* machine". That works, and it is how you keep three CTs from all showing each other's patients (see *Sending each scanner its own work*). It has one sharp edge:

> **A study with no station is invisible to a scanner that filters by station.** Not "shown anyway" — invisible. Measured: 65 X-ray studies scheduled for one day answered a modality-and-date query and returned **zero** to an X-ray scanner asking by its own AE title, because no station was mapped for that location and modality.

So the mapping and the scanner configuration are one job, not two. Either:

- every location and modality that scanner needs has a row in **Imaging Stations**, or
- that scanner filters by **modality and date** only.

`AutoTasks.out` counts the gap every cycle:

```
EmitWorklistFiles: published with NO station -- invisible to any scanner that
filters by station AE title   135   example location/modality   SiteC/US SiteA/DX ...
```

Those example pairs are exactly the rows missing from Imaging Stations.

6.6 A study is on the schedule but not in the worklist

In order of likelihood:

1. **Its revenue code has no modality.** By far the most common. The `no modality (not emitted)` count in the log is these.
2. **Its encounter is closed or inactive.** Both remove it deliberately.
3. **Its date is outside the window.** `MWL_WINDOW_DAYS` is one day either side of today by default.
4. **The scanner is filtering by a different modality** than the one the revenue code maps to — an ultrasound-guided procedure coded as a procedure rather than as ultrasound, for instance.

`AutoTasks.exe --mwl-run` settles 1-3 in a few seconds.

6.7 Things that look like faults and are not

What you see	Why it is correct
Files vanish from the worklist directory	AutoTasks owns that directory and sweeps it every cycle. That sweep is what withdraws cancelled studies
A file you copied in by hand disappears	Same sweep. Use `--mwl-selftest`, which is meant for this
`unchanged` is large and `written` is `0`	Correct on a quiet cycle. Files are rewritten only when something changed
Laboratory and office-visit orders are counted as "no modality"	They are not imaging, and belong in that count. It is not a number that must reach zero — only the imaging codes in it matter
The worklist directory contains a `.tmp` file for a moment	Files are written under a temporary name and renamed, so a scanner can never read a half-written one
Orthanc refuses a C-STORE	Deliberate. This server serves worklists and must never become an image archive
The worklist directory was emptied by hand, or restored from backup	Nothing to do. The generator compares what is on disk, not only what it remembers writing, so the files come back on the next cycle
A study that has already been scanned is still on the list	Expected — see *A finished study stays on the list*

7 Reference

7.1 Mp10 settings

Section	Entry	Default	What it does
GENERAL	MODALITY-WORKLIST	NO	Whether the `modalitywl` table exists. Read during a file check
AUTOTASKS	EXPORT_MWL	NO	Runs the generator every cycle
AUTOTASKS	MWL_SPOOL_PATH	*(blank)*	The worklist directory. Blank publishes nothing
AUTOTASKS	MWL_WINDOW_DAYS	1	Today +/- this many days
AUTOTASKS	MWL_UID_ROOT	*(blank)*	Leave blank unless your organisation owns a registered DICOM root. Never invent one
AUTOTASKS	MWL_SCP_PORT	4242	The DICOM server's port, for the watchdog. `0` switches the watchdog off
AUTOTASKS	MWL_SCP_AET	MP10MWL	The DICOM server's AE title, for the watchdog
AUTOTASKS	MWL_SCP_RESTART	YES	Whether the watchdog restarts the Orthanc service. `NO` = notices only
ERRORREPORTS	NOTIFY_EMAIL	*(blank)*	Who is told when the worklist server stops answering (and when an Mp10 program fails). Blank = nobody is mailed

7.2 Commands

Command	What it does
`AutoTasks.exe --mwl-run`	One worklist cycle, printed to the screen. Nothing else runs
`AutoTasks.exe --mwl-selftest <dir> [modality] [station AE]`	Publishes one fake study, with no database connection. Give it a station AE title to prove a station-filtering scanner
`AutoTasks.exe --mwl-probe [port] [AE title]`	Asks the DICOM server whether it is answering, with no database connection. Changes nothing
`AutoTasks.exe --mwl-watchdog`	The service's own server check, on demand: restarts the Orthanc service if it is not answering, and sends the notice
`Test-Mwl.ps1`	Reads both halves and reports what is wrong. Changes nothing
`Test-Mwl.ps1 -ExpectedSpoolPath <dir>`	Also confirms both halves use the same directory

Command	What it does
`Capture-MwlQuery.ps1`	Shows live what each scanner sends: its AE title, its query, how many entries it got. Ctrl+C to stop; the only thing it changes is the log level, and it puts that back
`Install-Mwl.ps1`	Installs or repairs the DICOM server. Safe to re-run, including while it is running
`Install-Mwl.ps1 -AcceptAnyCalledAet`	Answers whatever Called AE title a device uses — for legacy equipment whose worklist AE title is fixed
`Install-Mwl.ps1 -FilterByCallingAet`	A query that names no station gets only its caller's station — for scanners that cannot filter by their own AE title. Every scanner then needs its Imaging Stations row

7.3 What each worklist entry carries

The machine shows	Mp10 source
Patient name	Patient record, as `Family^Given`
Patient ID	Medical record number, padded to 8 digits — the same one your imaging system already knows
Date of birth, sex	Patient record
Accession number	The **encounter** number
Requested procedure ID	The **order** number — one study per order
Description	The item's description
Modality	The revenue code's modality
Scheduled date and time	The order's start date, else the encounter's admission date and time
Referring physician	The order's requesting physician, else the encounter's doctor
Scheduled station AE title	The scanner it is booked on, from Imaging Stations. Absent when that (location, modality) is not mapped
Station name and location	The same row's label and the encounter's site, for display
Study identifier	Issued once by Mp10 and never changed

The study identifier is worth understanding: it is issued when the study first appears on the worklist, stored, and reused for the life of that order. Cancel and reinstate a study and it keeps the same identifier, which is why images taken before and after never split into two studies.